

The “Difficulties” in Fermat’s Original Discourse on the Indecomposability of Powers Greater Than a Square: A Retrospect

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Abstract

We present a reconstruction of Pierre Fermat’s possible manuscript reasoning based on the two-power decomposition constraint and on the coefficient symmetry of the odd part of the binomial expansion.

Starting from a hypothetical positive natural solution of

$$x^n + y^n = z^n, \quad n > 2,$$

we use the symmetric parametrization

$$z = m^n + p^n, \quad x = m^n - p^n,$$

where m and p are first treated as positive real parameters. This gives

$$y^n = (m^n + p^n)^n - (m^n - p^n)^n.$$

The odd part of this binomial difference admits two natural real normalizations. The first extracts the linear binomial factor $n = C_n^1$:

$$y^n = 2n l^n.$$

The second extracts the total odd binomial coefficient mass. By the symmetry of Pascal’s triangle,

$$\sum_{r \text{ odd}} C_n^r = 2^{n-1},$$

and hence the second normalization has the form

$$y^n = 2^n q^n.$$

Here l and q are positive real residual scales. The symbol q denotes a scale parameter and must not be confused with a modulus in modular arithmetic.

Comparing the two representations gives the exact relation

$$l^n = q^n \frac{2^{n-1}}{n}.$$

Thus, within the two real normalizations of the same odd binomial core, the residual-scale equality

$$l^n = q^n$$

is not merely an additional numerical condition. It is equivalent to the equality between the linear coefficient n and the total odd coefficient mass 2^{n-1} :

$$l^n = q^n \iff n = 2^{n-1}.$$

The latter equation has exactly two positive integer solutions,

$$n \in \{1, 2\}.$$

Therefore the binomial coefficient symmetry gives a compatibility obstruction: the equality of the two residual scales is possible only in the linear and quadratic cases, and necessarily breaks for every exponent $n > 2$.

Consequently, if the internal two-power symmetry of a hypothetical positive integer solution is required to preserve the equality of the two residual scales, then every hypothetical solution with $n > 2$ leads to a contradiction.

The decomposition premise is kept as an explicit structural assumption over $\mathbb{N}$. The real parametrization is used as an algebraic reconstruction, while the integer specialization, parity conditions, and modular caveat are treated separately. In particular, modular congruences are not identified with integer equalities. The accompanying Coq formalization verifies the final exponential-linear obstruction

$$n = 2^{n-1} \implies n \in \{1, 2\},$$

and related bridge lemmas. In the present version, the role of binomial symmetry is sharpened: it explains why the residual-scale equality holds in the quadratic case and why it is incompatible with all exponents higher than the square.

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1 Introduction

The present work is the result of an attempted reconstruction of Fermat's original discourse along with an explanation of why he might have not written it down. The author had performed it within a two-time period of time— between 1990 and 1993 – trying to prove the theorem and the final revision in 2017-2026.¹ When completed, it did look like a proof of Fermat's epoch, as it only involved the knowledge and techniques available and utilised by Fermat's contemporary and pre-Fermat mathematical world.

Not to overburden this text with details of a real historical study, let us briefly recall the history of the conjecture. Around 1637, Fermat wrote his Last Theorem in the margin of his copy of the *Arithmetica* next to Diophantus' sum-of-squares problem [7]:

Cubum autem in duos cubos, aut quadratoquadratum in duos quadratoquadra-tos & generaliter nullam in infinitum ultra quadratum potestatem in duos eiusdem nominis fas est dividere cuius rei demonstrationem mirabilem sane detexi. Hanc marginis exiguitas non caperet.

Attempts to prove this conjecture employed a diverse range of methods. Early attempts, including Fermat's own proof for the case $n = 4$, utilized the **method of infinite descent**. A significant step forward was Leonhard Euler's proof for the case $n = 3$, which involved the use of **Gaussian integers** [13]. Subsequent efforts included approaches based on the work of Sophie Germain and other techniques, before Andrew Wiles finally presented a complete proof in 1995, drawing upon deep results from **the theory of modular forms and elliptic curves** [1–6].

Modern methods, such as the theory of modular forms, deal with **the transformation properties of specific curves over particular types of spaces** (e.g., rational numbers), highlighting the **stability of elliptic curves with respect to modular transformations**. While the contemporary formalizations of algebraic curves, spaces, transformations, and groups used in these methods were not present in Fermat's time, mathematicians of that era possessed their own approaches and intuitive understandings for studying the properties of natural numbers (and primes). (More on this can be found in the Conclusions section of this paper.)

Theorem 1 (Coefficient-symmetry obstruction for the two-power decomposition). *Assume that any hypothetical positive natural solution of the Fermat equation*

$$x^n + y^n = z^n$$

with $n > 2$ admits the real symmetric parametrization

$$z = m^n + p^n, \quad x = m^n - p^n,$$

¹ The work was mainly performed on the scholarship of the NRNU MEPhI student in 1990 – 93, final revision in 2017 – 2026.

with positive real parameters $m, p > 0$. Then

$$y^n = (m^n + p^n)^n - (m^n - p^n)^n.$$

Let the odd binomial core of this difference be represented in two natural ways by positive real residual scales $l, q > 0$:

$$(m^n + p^n)^n - (m^n - p^n)^n = 2n l^n,$$

and

$$(m^n + p^n)^n - (m^n - p^n)^n = 2^n q^n.$$

Here l is the residual scale obtained after extracting the linear binomial factor $n = C_n^1$, while q is the residual scale obtained after extracting the total odd binomial coefficient mass 2^{n-1} . In this theorem, q is a positive real scale and must not be confused with a modulus in modular arithmetic.

Then the comparison of these two natural normalizations gives the exact compatibility relation

$$l^n = q^n \frac{2^{n-1}}{n}.$$

Consequently, the residual-scale equality

$$l^n = q^n$$

is possible if and only if

$$n = 2^{n-1}.$$

Since this equation has only the positive integer solutions $n = 1$ and $n = 2$, the residual-scale equality is incompatible with every exponent $n > 2$.

Therefore, if the internal two-power symmetry of a hypothetical positive integer solution is required to preserve the equality of the two residual scales, then the Fermat equation

$$x^n + y^n = z^n$$

has no positive natural solutions for any $n > 2$.

Proof sketch. Given a putative positive natural solution at a fixed exponent $n > 2$, the two-power decomposition gives the binomial identity

$$y^n = (m^n + p^n)^n - (m^n - p^n)^n.$$

The odd part of the binomial expansion admits two real normalizations:

$$y^n = 2n l^n$$

and

$$y^n = 2^n q^n.$$

The first normalization extracts the linear coefficient $n = C_n^1$. The second normalization extracts the full odd coefficient mass. By the symmetry of Pascal's triangle,

$$\sum_{r \text{ odd}} C_n^r = 2^{n-1}.$$

Thus the two extracted coefficient masses are n and 2^{n-1} .

Equating the two expressions for y^n gives

$$2n l^n = 2^n q^n.$$

Since $2n > 0$, division by $2n$ gives the exact relation

$$l^n = q^n \frac{2^{n-1}}{n}.$$

This relation is not an additional hypothesis; it follows from the two normalizations of the same odd binomial core.

Now suppose that the two residual scales are required to coincide:

$$l^n = q^n.$$

Substituting this into the exact relation gives

$$q^n = q^n \frac{2^{n-1}}{n}.$$

Since $q > 0$, we have $q^n > 0$, and cancellation of q^n is legitimate. Hence

$$1 = \frac{2^{n-1}}{n},$$

or equivalently

$$n = 2^{n-1}.$$

Conversely, if

$$n = 2^{n-1},$$

then the exact relation

$$l^n = q^n \frac{2^{n-1}}{n}$$

immediately gives

$$l^n = q^n.$$

Therefore, under the two natural real normalizations of the same odd binomial core, we have the equivalence

$$l^n = q^n \iff n = 2^{n-1}.$$

The equation

$$n = 2^{n-1}$$

has exactly two positive integer solutions:

$$n \in \{1, 2\}.$$

Indeed, it holds for $n = 1$ and $n = 2$, while for every integer $n > 2$ the exponential term 2^{n-1} strictly dominates the linear term n :

$$2^{n-1} > n.$$

Therefore the equality of the two residual scales is impossible for every exponent higher than the square.

Thus the binomial coefficient symmetry does not merely motivate the second normalization. It gives a precise compatibility obstruction: the equality

$$l^n = q^n$$

is compatible with the two binomial normalizations only in the cases

$$n = 1 \quad \text{or} \quad n = 2.$$

Consequently, if the internal two-power symmetry of a hypothetical positive integer solution is required to preserve this equality of residual scales, then a putative solution with $n > 2$ leads to a contradiction.

The elementary growth comparison

$$n = 2^{n-1} \implies n \in \{1, 2\}$$

is the same final exponential-linear obstruction that is machine-checkable in the accompanying Coq file currently named `GlobalNormalization.v`. In the present version, the role of the binomial symmetry is sharpened: it explains why the residual-scale equality holds in the quadratic case and why it necessarily breaks for every exponent $n > 2$.

2 Possible Proof

The statement of the theorem is rather straightforward and as follows:

Neither a cube for two cubes, nor a biquadrate or two biquadrates, and generally no power greater than two can be decomposed into two powers of the same grade. In other words, the equation

$$x^n + y^n = z^n$$

has no solutions in natural numbers if n is an integer greater than 2.

Therefore, first

- 1). Let's write down the theorem

$$x^n + y^n = z^n \tag{2.1}$$

- 2). Let's rewrite (2.1)

$$z^n - x^n = y^n \tag{2.2}$$

- 3). Let's perform the transformation of (2.2) to

$$(m^n + p^n)^n - (m^n - p^n)^n = y^n, \tag{2.3}$$

where $n \in \mathbb{N}$, m, p are arbitrary numbers (not necessarily integers; signs arbitrary), and $z, x \in \mathbb{N}$ satisfy

$$\begin{cases} m^n + p^n = z, \\ m^n - p^n = x. \end{cases} \tag{2.4}$$

Raising both identities to the n -th power yields

$$\begin{cases} (m^n + p^n)^n = z^n, \\ (m^n - p^n)^n = x^n. \end{cases} \tag{2.5}$$

Thus (2.4) implies (2.5).

Solving (2.4). Adding and subtracting the equations gives

$$2m^n = z + x, \quad 2p^n = z - x,$$

so formally

$$m = \sqrt[n]{\frac{z+x}{2}}, \quad p = \sqrt[n]{\frac{z-x}{2}}.$$

Domain note. Over $\mathbb{R}$: for odd n the roots are unique; for even n we need $(z \pm x)/2 \geq 0$ and there is a sign choice $m = \pm((z+x)/2)^{1/n}$, $p = \pm((z-x)/2)^{1/n}$. Over $\mathbb{C}$: there are n branches for the n -th root; once a branch is fixed, the reconstruction is consistent.

Equivalence of (2.4)–(2.5). **Forward** (2.4) $\Rightarrow$ (2.5). Immediate by raising to the n -th power.

Reverse (2.5) $\Rightarrow$ (2.4). Assume $z, x \in \mathbb{N}$ and

$$z^n = (m^n + p^n)^n, \quad x^n = (m^n - p^n)^n.$$

Taking n -th roots in the same domain as $m^n \pm p^n$ (principal root over $\mathbb{R}_{\geq 0}$, or a fixed branch over $\mathbb{C}$) gives

$$z = m^n + p^n, \quad x = m^n - p^n,$$

i.e. (2.4). For even n one fixes consistent signs as above. Hence, (2.4) and (2.5) are equivalent in the chosen number domain once the root/branch convention is fixed.

Proposition 1 (Integer case: necessary & sufficient conditions). *If, in addition, one requires $m, p \in \mathbb{Z}$, then necessarily*

$$z \pm x \text{ are even,} \quad \frac{z+x}{2} = m^n, \quad \frac{z-x}{2} = p^n.$$

Conversely, if $z \pm x$ are even and both halves are perfect n -th powers in $\mathbb{Z}$, then the reconstructed m, p are integers (up to signs for even n).

Remark 1. *This separates the general real/complex reconstruction (no parity obstruction) from the integer reconstruction, where parity and perfect-power constraints are essential.*

Example (real domain). For $n = 2$, $m = 3$, $p = 2$:

$$z = 3^2 + 2^2 = 13, \quad x = 3^2 - 2^2 = 5,$$

hence

$$z^2 = 169, \quad x^2 = 25,$$

and reversing via $(z \pm x)/2$ returns $m = 3$, $p = 2$.

Counterexample (integer case). For $n = 3$, $z = 2$, $x = 1$ we have $(z+x)/2 = 3/2$, which is not a perfect cube in $\mathbb{Z}$; hence no integer m, p satisfy (2.4), although real m, p exist.

Modular remark. The present work does not construct a proof by means of modular arithmetic and does not claim the absence of solutions to the corresponding congruences over finite fields. The scheme considered here concerns equalities over the natural, integer, rational, and real numbers, not equations over a finite field $\mathbb{F}_q$.

In particular, from an integer equality

$$x^n + y^n = z^n$$

one indeed obtains the congruence modulo any q :

$$x^n + y^n \equiv z^n \pmod{q}.$$

However, the converse implication is not valid in general: from the congruence

$$x^n + y^n \equiv z^n \pmod{q}$$

one cannot infer the equality

$$x^n + y^n = z^n$$

over the integers.

In other words, a modular congruence only means that there exists an integer t such that

$$z^n = x^n + y^n + qt.$$

An integer solution of Fermat's equation corresponds to the special case

$$t = 0.$$

Therefore, the existence of solutions to the congruences

$$x^n + y^n \equiv z^n \pmod{q}$$

for $n > 2$ is not a counterexample to Fermat's Last Theorem and does not disprove the real parametrization used in the present work.

If one specifically passes to a modulus q , then the parameters must be specified separately at the modular level. For example, instead of the real equalities

$$z = m^n + p^n, \quad x = m^n - p^n$$

one should separately fix the residues

$$m^n \equiv A \pmod{q}, \quad p^n \equiv B \pmod{q},$$

after which the corresponding modular relations take the form

$$z \equiv A + B \pmod{q}, \quad x \equiv A - B \pmod{q}.$$

Thus, the real parametrization and modular arithmetic belong to different levels of description. In this article, the first level is used: equalities over $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, and $\mathbb{R}$, rather than a separate theory of solutions over finite fields.

- 4). Let's ask ourselves the question: what is the number y ? Is it positive integer or not? If it is not positive integer, then under what conditions will it be natural number? Does its naturalness depend on the degree of n ?

From (2.3) we have the difference:

$$y^n = z^n - x^n = (m^n + p^n)^n - (m^n - p^n)^n = \quad (2.6)$$

that can be expanded or decomposed into a sum according to Newton's binomial [8, 9]:

$$\begin{aligned} &= [(m^n)^n + C_n^1(m^n)^{n-1}p^n + C_n^2(m^n)^{n-2}(p^n)^2 + \dots + C_n^{n-1}m^n(p^n)^{n-1} + (p^n)^n] - \\ &- [(m^n)^n - C_n^1(m^n)^{n-1}p^n + C_n^2(m^n)^{n-2}(p^n)^2 \pm \dots \pm C_n^{n-1}m^n(p^n)^{n-1} \pm (p^n)^n] = \\ &= 2C_n^1(m^n)^{n-1}p^n + 2C_n^3(m^n)^{n-3}(p^n)^3 + \dots + 2C_n^k(m^n)^{n-k}(p^n)^k + \{2C_n^n(p^n)^n\} = \\ &= 2 \sum_{i=0}^k C_n^{(2i+1)}(m^n)^{n-(2i+1)}(p^n)^{(2i+1)} \end{aligned}$$

with $k = (n-1)/2$ if n is odd and $k = (n-2)/2$ if n is even.

Rewrite then (2.6) as

$$\begin{aligned} &z^n = x^n + y^n, \\ &\text{where } x, y, z \text{ are } \begin{cases} z = m^n + p^n \\ x = m^n - p^n \\ y = \sqrt[n]{2} \left[\sum_{i=0}^k C_n^{(2i+1)}(m^n)^{n-(2i+1)}(p^n)^{(2i+1)} \right]^{1/n} \end{cases} \end{aligned}$$

with $k = (n-1)/2$ if n is odd and $k = (n-2)/2$ if n is even;

we know that in common case for $n = 2$ we have

$$\text{where } x, y, z \text{ are given by } \begin{cases} z = r \cdot (m^2 + p^2), \\ x = r \cdot (m^2 - p^2), \\ y = r \cdot 2mp. \end{cases} \quad (2.7)$$

but in our case, we omit the factor r – is some positive ($r > 0$) integer constant since it is reduced in our calculations. Therefore, we made the transformation (2.3) in order to take into account this case ($n=2$), known since the time of Pythagoras, since the general case should include a particular solution as a subset.

- 5). scrutinise now the y :

$$y = \sqrt[n]{2} \left[\sum_{i=0}^k C_n^{(2i+1)}(m^n)^{n-(2i+1)}(p^n)^{(2i+1)} \right]^{1/n}. \quad (2.8)$$

In order for the y to maybe a positive integer, $\sqrt[n]{2}$ must leave, since for $n > 1$ $\sqrt[n]{2}$ is an irrational number (see Appendix A). It is thus necessary that the expression

$$\left[\sum_{i=0}^k C_n^{(2i+1)} (m^n)^{n-(2i+1)} (p^n)^{(2i+1)} \right]^{1/n} \quad (2.9)$$

contain some common factor that destroys the radical expression $\sqrt[n]{2}$, let's find out what it is. Otherwise, y is not a positive integer due to the presence of $\sqrt[n]{2}$. Consider now what largest divisor this sum may contain and what it is equal to:

$$\begin{aligned} & \left[\sum_{i=0}^k C_n^{(2i+1)} (m^n)^{n-(2i+1)} (p^n)^{(2i+1)} \right] = \\ &= C_n^1 (m^n)^{n-1} p^n + C_n^3 (m^n)^{n-3} (p^n)^3 + \dots + C_n^k (m^n)^{n-k} (p^n)^k + \{C_n^n (p^n)^n\} = \\ &= n(m^n)^{n-1} p^n + \frac{n(n-1)(n-2)}{3!} (m^n)^{n-3} p^3 + \dots + \\ &+ \frac{n(n-1)\dots(n-k+1)}{k!} (m^n)^{n-k} (p^n)^k + \left\{ \frac{n(n-1)\dots 2 \cdot 1}{n!} (p^n)^n \right\} = \\ &= n \cdot \left[(m^n)^{n-1} p^n + \frac{(n-1)(n-2)}{3!} (m^n)^{n-3} p^3 + \dots + \right. \\ &\left. + \frac{(n-1)\dots(n-k+1)}{k!} (m^n)^{n-k} (p^n)^k + \left\{ \frac{(n-1)\dots 2 \cdot 1}{n!} (p^n)^n \right\} \right] = n \cdot l^n \end{aligned} \quad (2.10)$$

with $k = (n-1)/2$ if n is odd and $k = (n-2)/2$ if n is even,

Hence the conclusion follows: the expression admits a common factor n in the real algebraic factorization. Here l denotes a scaling constant about which nothing is assumed a priori; in the general case it may be real, and in special arithmetic cases it may be integer. It represents the remaining part of the radical after extracting the common factor n .

Since each term in the expression can be written with a common factor n , the whole sum can be factored over the real domain in the form

$$\sum_{i=0}^k C_n^{(2i+1)} (m^n)^{n-(2i+1)} (p^n)^{2i+1} = n l^n.$$

Here l is a real scaling parameter defined by this factorization. Thus, this step should be understood as extracting a common real factor n , not as claiming integer divisibility of the whole expression by n .

To show the uniqueness of the decomposition of expression (2.10), we can use the following considerations:

- (a) Degree n is a fixed positive integer.
- (b) Exponentiation is an unambiguous operation; for any number, there is a unique value for the degree.
- (c) Addition and subtraction of real numbers are commutative operations, and the result does not depend on the order of actions.

Based on these properties, it can be argued that for fixed m , p and n there is a single result of calculating the expression (2.10). Rearranging the members will not affect the final answer.

We can see from (2.8)–(2.10) (where l is a constant depending on the fixed values of m , p , and n , and about which nothing else is assumed at this stage)

$$y = \sqrt[n]{2 \cdot n} \cdot l. \quad (2.11)$$

As a result, from (2.6)–(2.11) we get

$$z^n - x^n = (m^n + p^n)^n - (m^n - p^n)^n = y^n. \quad (2.12)$$

Substituting (2.11) into (2.12), we have

$$(m^n + p^n)^n - (m^n - p^n)^n = 2n l^n. \quad (2.13)$$

The second symmetric scale and the coefficient-symmetry obstruction.

Let us denote the odd binomial core by

$$S_n(m, p) := \sum_{i=0}^k C_n^{(2i+1)} (m^n)^{n-(2i+1)} (p^n)^{2i+1},$$

where $k = (n-1)/2$ if n is odd and $k = (n-2)/2$ if n is even. Then the binomial expansion gives

$$(m^n + p^n)^n - (m^n - p^n)^n = 2S_n(m, p).$$

From the previous real factorization of the odd binomial core we have

$$S_n(m, p) = n l^n,$$

where $l > 0$ is a real residual scale. Hence

$$(m^n + p^n)^n - (m^n - p^n)^n = 2n l^n. \quad (2.13a)$$

On the other hand, by the symmetry of Pascal's triangle,

$$C_n^r = C_n^{n-r},$$

the total mass of the odd binomial coefficients is equal to the total mass of the even binomial coefficients and is therefore one half of the full binomial mass:

$$\sum_{i=0}^k C_n^{(2i+1)} = 2^{n-1}.$$

This identity is the precise combinatorial place where the binary symmetry of the binomial expansion enters the argument.

Therefore, we may introduce a second positive real residual scale $q > 0$, obtained after extracting the total odd coefficient mass 2^{n-1} :

$$S_n(m, p) = 2^{n-1} q^n.$$

Equivalently,

$$q^n = \frac{1}{2^{n-1}} \sum_{i=0}^k C_n^{(2i+1)} (m^n)^{n-(2i+1)} (p^n)^{2i+1}.$$

This is not a claim that the weighted binomial core is equal to the pure sum of its coefficients. Rather, it is the definition of the second real residual scale q after the coefficient mass 2^{n-1} has been extracted.

Thus the same difference of powers also admits the representation

$$(m^n + p^n)^n - (m^n - p^n)^n = 2^n q^n. \quad (2.14)$$

Comparing (2.13a) and (2.14), we obtain

$$2n l^n = 2^n q^n.$$

After division by $2n > 0$, this gives the exact relation

$$l^n = q^n \frac{2^{n-1}}{n}. \quad (2.15)$$

What the binomial symmetry really proves. The two quantities l and q arise from two natural normalizations of the same odd binomial core:

$$S_n(m, p) = n l^n, \quad S_n(m, p) = 2^{n-1} q^n.$$

The first normalization extracts the linear coefficient $n = C_n^1$. The second normalization extracts the total odd binomial mass 2^{n-1} , which is forced by the symmetry of the binomial coefficients. Consequently, the equality of the residual scales

$$l^n = q^n$$

is not a separate numerical accident. It is equivalent, through (2.15), to the equality between the linear coefficient and the total odd coefficient mass:

$$n = 2^{n-1}.$$

Indeed, if $l^n = q^n$, then (2.15) gives

$$q^n = q^n \frac{2^{n-1}}{n}.$$

Since $q > 0$, we have $q^n > 0$, so cancellation of q^n is legitimate, and hence

$$1 = \frac{2^{n-1}}{n}.$$

Equivalently,

$$n = 2^{n-1}. \tag{2.16}$$

Conversely, if $n = 2^{n-1}$, then (2.15) immediately gives

$$l^n = q^n.$$

Thus, under the two real normalizations of the same odd binomial core, we have the exact equivalence

$$l^n = q^n \iff n = 2^{n-1}. \tag{2.17}$$

The quadratic case. For $n = 2$, the relevant row of Pascal's triangle is

$$1, \quad 2, \quad 1.$$

There is only one odd-position binomial coefficient in the difference of the two symmetric powers, namely

$$C_2^1 = 2.$$

The linear coefficient extracted in the first normalization is

$$n = 2,$$

and the total odd coefficient mass extracted in the second normalization is

$$2^{2-1} = 2.$$

Thus the two extracted coefficient masses coincide:

$$n = 2^{n-1}.$$

Consequently, the two residual scales also coincide:

$$l^2 = q^2.$$

This is exactly the quadratic, Pythagorean case.

Why higher exponents break the equality of scales. For $n = 3$, the odd part of the binomial expansion contains more than one odd coefficient:

$$C_3^1 = 3, \quad C_3^3 = 1,$$

so the total odd coefficient mass is

$$2^{3-1} = 4.$$

The first normalization extracts 3, whereas the second normalization extracts 4. Hence

$$l^3 = q^3 \frac{4}{3},$$

and therefore

$$l^3 \neq q^3.$$

For $n = 4$, the odd coefficients are

$$C_4^1 = 4, \quad C_4^3 = 4,$$

so their total mass is

$$2^{4-1} = 8.$$

The first normalization extracts 4, while the second extracts 8. Hence

$$l^4 = 2q^4,$$

and again

$$l^4 \neq q^4.$$

In general, for every integer $n > 2$,

$$2^{n-1} > n.$$

Therefore, by (2.15),

$$l^n = q^n \frac{2^{n-1}}{n} > q^n. \quad (2.18)$$

Thus the binomial coefficient symmetry does not prove that $l^n = q^n$ for all exponents. It proves the sharper statement that the equality of the two residual scales is compatible with the binomial coefficient structure only in the cases

$$n = 1 \quad \text{or} \quad n = 2.$$

Analysis of the equation $n = 2^{n-1}$.

The equation (2.16) is equivalent to

$$2^n = 2n.$$

It has exactly two positive integer solutions:

$$n = 1, \quad n = 2.$$

Indeed,

$$2^{1-1} = 1, \quad 2^{2-1} = 2.$$

For every integer $n > 2$, the exponential term 2^{n-1} strictly dominates the linear term n . Hence

$$2^{n-1} > n \quad \text{for all } n > 2.$$

Therefore, equation (2.16) is impossible for every exponent higher than the square.

Conclusion: coefficient symmetry as a compatibility obstruction.

The binomial symmetry of Pascal's triangle does not merely motivate the second normalization. It gives a precise compatibility test for the equality of the two residual scales. Under the two normalizations

$$S_n(m, p) = n l^n, \quad S_n(m, p) = 2^{n-1} q^n,$$

the condition

$$l^n = q^n$$

is equivalent to

$$n = 2^{n-1}.$$

Since this equation has only the positive integer solutions

$$n \in \{1, 2\},$$

the equality of the two residual scales is possible only in the linear and quadratic cases.

Thus, in Version 15, the role of the binomial symmetry is sharpened: it explains why the residual scale equality holds in the quadratic case and why it necessarily breaks for every exponent $n > 2$. If the internal two-power symmetry of a hypothetical positive integer solution is required to preserve equality of the two residual scales, then every exponent higher than the square is excluded.

6.) Verification (without considering the total multiplier r from (2.7), which has been reduced)

(a) Consider the case $n = 1$

$$\begin{aligned} z &= m + p \\ x &= m - p \\ y &= 2 \cdot [C_1^1(m^1)^{1-(2 \cdot 0+1)}(p^1)^{(2 \cdot 0+1)}] = 2[1 \cdot 1 \cdot p] = 2p \end{aligned}$$

that is, for the case $n = 1$, we have a solution in natural numbers x, y, z , if m and p - are positive integer numbers and $m > p$.

(b) Consider the case $n = 2$

$$\begin{aligned} z &= m^2 + p^2 \\ x &= m^2 - p^2 \\ y &= \sqrt{2} \cdot [C_1^2(m^2)^{2-(2 \cdot 0+1)}(p^2)^{(2 \cdot 0+1)}]^{1/2} = \sqrt{2} \cdot [2m^2 p^2]^{1/2} = 2mp \end{aligned}$$

that is, for the case $n = 2$, we also have a solution in positive integer numbers x, y, z , if m and p are positive integer numbers and $m > p$, or according to [10], if $m > p$, and m, p are such non-integer numbers that combinations with them will give the positive integer numbers x, y, z , for example: $m = 3/\sqrt{2}$, $p = 1/\sqrt{2}$, in this case, we also get the classical Pythagorean triple:

$$\begin{aligned} x &= m^2 - p^2 = \left(\frac{3}{\sqrt{2}}\right)^2 - \left(\frac{1}{\sqrt{2}}\right)^2 = \frac{9}{2} - \frac{1}{2} = \frac{8}{2} = 4 \\ y &= 2 \cdot m \cdot p = 2 \cdot \frac{3}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} = 3 \\ z &= m^2 + p^2 = \left(\frac{3}{\sqrt{2}}\right)^2 + \left(\frac{1}{\sqrt{2}}\right)^2 = \frac{9}{2} + \frac{1}{2} = 5 \end{aligned}$$

We have obtained the classical Pythagorean triple and can see that this formula works.

- 7). Let us now consider the more general case where r is not reduced to proceed to final formulas like (2.7). If r is a positive integer number, then all the above applies. But according to [10], r can be a rational number. For example, let's say $r = 0.5$:

- (a) Consider the case of $n = 1$, $m = 4$, and $p = 2$

$$x = 0.5 \cdot (m - p) = 0.5 \cdot (4 - 2) = 1$$

$$z = 0.5 \cdot (m + p) = 0.5 \cdot (4 + 2) = 3$$

$$y = 0.5 \cdot (2 \cdot p) = 0.5 \cdot (2 \cdot 2) = 2$$

We see positive integer numbers. That is,, m , p must be such that 7(a) is performed similarly. In this case, we can see that the formula 6(a) works.

- (b) Consider the case of $n = 2$, $m = 2 \cdot \sqrt{2}$, $p = \sqrt{2}$,

$$x = 0.5 \cdot (m^2 - p^2) = 0.5 \cdot \left((2 \cdot \sqrt{2})^2 - (\sqrt{2})^2 \right) = 3$$

$$z = 0.5 \cdot (m^2 + p^2) = 0.5 \cdot \left((2 \cdot \sqrt{2})^2 + (\sqrt{2})^2 \right) = 5$$

$$y = 0.5 \cdot (2 \cdot m \cdot p) = 0.5 \cdot (2 \cdot (2 \cdot \sqrt{2}) \cdot (\sqrt{2})) = 4$$

We see positive integer numbers. That is,, m , p must be such that 7(b) holds similarly. In this case, we can see that the formula 6(b) works.

This preserves the generality of solutions for $n = 2$, as seen in the Pythagorean triple example.

- 8). Therefore, the cases $n = 1$ and $n = 2$ **exhaust all possible integer solutions**, which is consistent with the theorem.

The test showed that for $n=1$ or for $n=2$, in all the cases considered, we have solutions of the equation $x^n + y^n = z^n$ in the positive integer numbers x , y , z

- 9). **the equation $x^n + y^n = z^n$ has roots in the positive integer numbers x , y , z only for $n = 1$ and for $n = 2$**

This completes the conditional argument based on the residual-scale compatibility obstruction.

3 Remark and corollaries

REMARK. Note that the expression (2.8) can be simplified, namely

$$\begin{aligned} y &= \sqrt[n]{2} \left[\sum_{i=0}^k C_n^{(2i+1)} (m^n)^{n-(2i+1)} (p^n)^{(2i+1)} \right]^{1/n} = \\ &= \sqrt[n]{2} \left[\sum_{i=0}^k C_n^{(2i+1)} \frac{(m^n)^n}{(m^n)^{2i} m^n} (p^n)^{2i} p^n \right]^{1/n} = \\ &= \sqrt[n]{2} m^n \frac{1}{m} p \left[\sum_{i=0}^k C_n^{(2i+1)} \left(\frac{p}{m} \right)^{n2i} \right]^{1/n} = \\ &= \sqrt[n]{2} m^{n-1} p \left[\sum_{i=0}^k C_n^{(2i+1)} \left(\frac{p}{m} \right)^{2in} \right]^{1/n} \end{aligned} \tag{3.1}$$

with $k = (n - 1)/2$ if n is odd and $k = (n - 2)/2$ if n is even.

COROLLARY 1. Consider the case $m = p$, then from the expression (3.1) it can be derived that

$$\begin{aligned} x &= 0 \\ z &= m^n + m^n = 2m^n \\ y &= \sqrt[n]{2} m^{n-1} m \left[\sum_{i=0}^k C_n^{(2i+1)} \left(\frac{m}{m} \right)^{2in} \right]^{1/n} = \sqrt[n]{2} m^n \left[\sum_{i=0}^k C_n^{(2i+1)} \right]^{1/n} \end{aligned}$$

with $k = (n - 1)/2$ if n is odd and $k = (n - 2)/2$ if n is even.

It is obvious that $y = z$

$$\sqrt[n]{2}m^n \left[\sum_{i=0}^k C_n^{(2i+1)} \right]^{1/n} = 2m^n$$

$$\sqrt[n]{2} \left[\sum_{i=0}^k C_n^{(2i+1)} \right]^{1/n} = 2,$$

whence

$$\sum_{i=0}^k C_n^{(2i+1)} = 2^{n-1} \quad (3.2)$$

with $k = (n-1)/2$ if n is odd and $k = (n-2)/2$ if n is even.

COROLLARY 2. *Based on (3.2), the sum of even combinations can be calculated.*

Consider Pascal's triangle [7]:

$$\begin{array}{ccccccc} & & 1 & & 1 & & \\ & & & 1 & & 2 & & 1 \\ & & & & 1 & & 3 & & 3 & & 1 \\ & & & & & 1 & & 4 & & 6 & & 4 & & 1 \\ & & & & & & 1 & & 5 & & 10 & & 10 & & 5 & & 1 \end{array}$$

Similarly to the above, it is concluded that

$$\sum_{j=0}^s C_n^{2j} = 2^{n-1} \quad (3.3)$$

with $k = (n-1)/2$ if n is odd and $k = (n-2)/2$ if n is even.

PROOF.

Expand

$$(m^n + p^n)^n + (m^n - p^n)^n =$$

into a binomial [8, 9]:

$$\begin{aligned} &= [(m^n)^n + C_n^1(m^n)^{n-1}p^n + C_n^2(m^n)^{n-2}(p^n)^2 + \dots + C_n^{n-1}m^n(p^n)^{n-1} + (p^n)^n] + \\ &+ [(m^n)^n - C_n^1(m^n)^{n-1}p^n + C_n^2(m^n)^{n-2}(p^n)^2 \pm \dots \pm C_n^{n-1}m^n(p^n)^{n-1} \pm (p^n)^n] = \\ &= 2C_n^0(m^n)^n + 2C_n^2(m^n)^{n-2}(p^n)^2 + \dots + 2C_n^k(m^n)^{n-k}(p^n)^k + \{2C_n^n(p^n)^n\} = \\ &= 2 \sum_{j=0}^s C_n^{2j} (m^n)^{n-2j} (p^n)^{2j} \end{aligned}$$

with $s = (n-1)/2$ if n is odd and $s = n/2$ if n is even.

If $m = p$

$$(p^n + p^n)^n + (p^n - p^n)^n = 2 \sum_{j=0}^s C_n^{2j} (p^n)^{n-2j} (p^n)^{2j}$$

$$(2p^n)^n = 2 \sum_{j=0}^s C_n^{2j} \frac{(p^n)^n}{(p^n)^{2j}} (p^n)^{2j}$$

$$2^n (p^n)^n = 2 (p^n)^n \sum_{j=0}^s C_n^{2j}$$

$$2^{n-1} = \sum_{j=0}^s C_n^{2j}$$

with $s = (n-1)/2$ if n is odd and $s = n/2$ if n is even.

Corollary 2 is proved.

COROLLARY 3. *Analysing (3.2) and (3.3), it can be concluded that*

$$\sum_{i=0}^k C_n^{(2i+1)} = \sum_{j=0}^s C_n^{2j} \quad (3.4)$$

with $k = (n-1)/2$, $s = (n-1)/2$ if n is odd and $k = (n-2)/2$, $s = n/2$ if n is even. Why such borders?

Type of number	Condition	Last constant in equation
Odd	$k = 2((n-1)/2)+1 = n$	C_n^n
	$s = 2((n-1)/2) = n-1$	C_n^{n-1}
Even	$k = 2((n-2)/2)+1 = n-1$	C_n^{n-1}
	$s = 2(n/2) = n$	C_n^n

Table 1: Boundaries of binomial coefficients

Conclusion: the sum of even coefficients is equal to the sum of odd ones and is equal to 2^{n-1} , therefore from (3.4) we have

$$\sum_{r=0}^n C_n^r = \sum_{i=0}^k C_n^{(2i+1)} + \sum_{j=0}^s C_n^{2j} = 2 \cdot 2^{n-1} = 2^n \quad (3.5)$$

with $k = (n-1)/2$, $s = (n-1)/2$ if n is odd and $k = (n-2)/2$, $s = n/2$ if n is even.

4 Conclusion

The “difficulties” were for Fermat the lengthiness of the run of his deductions *put in writing*, since in the first half of the seventeenth century mathematical notation was still far from its present concise and diverse form, and many actions had to be written down *in words*. Besides, a purely mathematical challenge was that he had to operate with notions of binomial expansion and logarithmic comparison which were then still new, only emerging for systematic use and having to be learnt almost “on the fly”.

As mentioned in the introduction, the elementary algebraic part of the present reconstruction is accessible at the level of early university mathematics. This should be understood only as a comment on the elementary character of the reconstructed algebraic language, not as a comparison of mathematical strength with the modern proof of Fermat’s Last Theorem.

Andrew Wiles’s proof, completed with Richard Taylor, belongs to a very different modern framework involving modular forms, elliptic curves, Galois representations, and deformation theory. The present reconstruction does not compete with that proof. Rather, it explores whether a much more elementary algebraic line of thought could plausibly explain the kind of reasoning Fermat may have had in mind.

Fermat was, plausibly, “playing” with the new algebraic ideas of his time, decomposing powers of sums and differences and observing the special role of the binary structure. In the reconstruction proposed here, the decisive point is expressed not as an external normalization, but as a compatibility obstruction between two natural real normalizations of the same odd binomial core.

Indeed, the odd binomial core may be written in two ways:

$$S_n(m, p) = n l^n, \quad S_n(m, p) = 2^{n-1} q^n.$$

Equivalently, for the full difference of powers, this gives

$$y^n = 2n l^n, \quad y^n = 2^n q^n.$$

Comparing these two representations yields the exact relation

$$l^n = q^n \frac{2^{n-1}}{n}.$$

Therefore, under these two normalizations, the equality of the residual scales is possible if and only if

$$l^n = q^n \iff n = 2^{n-1}.$$

This is the coefficient-symmetry compatibility obstruction of the present version.

The elementary exponential-linear equation

$$n = 2^{n-1}$$

has positive integer solutions only for

$$n = 1 \quad \text{and} \quad n = 2.$$

Thus, the equality of the two residual scales is aligned with the linear and quadratic cases and is incompatible with every exponent higher than the square. In particular, if the internal two-power symmetry of a hypothetical positive integer solution is required to preserve the equality

$$l^n = q^n,$$

then the equation

$$x^n + y^n = z^n$$

cannot have positive integer solutions for any $n > 2$.

This is the sharpened role of the binomial symmetry in Version 15: it does not merely motivate a scale condition, but shows that the residual-scale equality is compatible with the binomial coefficient structure only in the cases $n = 1$ and $n = 2$. For every $n > 2$, the total odd binomial coefficient mass 2^{n-1} strictly exceeds the linear coefficient n , and the two residual scales necessarily separate.

He would first have had to introduce and explain these algebraic ideas in order to make his finding fully intelligible. One can imagine *how much room it would take to put down all the deliberations* that had led him to his discovery *on the margin of a book, without the symbolic notation available to a contemporary mathematician*.

Why Pierre Fermat did not write down all those ideas in a dedicated document is a separate historical question. It may turn out that he did write such a document, which was afterwards lost, or that it has survived to this day hidden in an archive, a library, or some unrealised private custody.

The author requests the mathematical community to look critically at the deliberations set forth above and to return their assessment.

This paper has been published as a preprint on the ResearchGate platform and on the OSF platform at the following links [11, 12].

The Coq formalization is available in the repository FLT-Coq on GitHub [14].

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A Appendix A. Proof of the Irrationality of $\sqrt[n]{2}$ for $n \geq 2$

A.1 Theorem: The number $\sqrt[n]{2}$ is irrational for any integer $n \geq 2$.

A.2 Proof:

We will prove by contradiction. Assume that $\sqrt[n]{2}$ is a rational number. This means it can be represented as an irreducible fraction $\frac{p}{q}$, where p and q are integers, $q \neq 0$, and p and q have no common divisors other than 1.

Thus, we have:

$$\sqrt[n]{2} = \frac{p}{q}$$

Raise both sides of the equation to the power of n :

$$(\sqrt[n]{2})^n = \left(\frac{p}{q}\right)^n$$

$$2 = \frac{p^n}{q^n}$$

Multiply both sides by q^n :

$$2q^n = p^n$$

From this equation, we see that p^n is an even number, since it is equal to $2q^n$. If p^n is even, then p must also be even (if p were odd, then p^n would also be odd).

Since p is even, we can write it as $p = 2k$, where k is some integer. Substitute this expression for p into the equation $2q^n = p^n$:

$$2q^n = (2k)^n$$

$$2q^n = 2^n k^n$$

Divide both sides of the equation by 2 (which is permissible since $n \geq 2$, and therefore 2^n is divisible by 2):

$$q^n = 2^{n-1} k^n$$

Since $n \geq 2$, then $n - 1 \geq 1$. From the last equation, we see that q^n is an even number, since it is equal to $2^{n-1} k^n$, where 2^{n-1} is an even factor. If q^n is even, then q must also be even.

Thus, we have reached the conclusion that both p and q are even numbers. This means they have a common divisor of 2. However, at the beginning of the proof, we assumed that the fraction $\frac{p}{q}$ was irreducible, meaning p and q have no common divisors other than 1.

The resulting contradiction indicates that our initial assumption about the rationality of $\sqrt[n]{2}$ is incorrect.

A.3 Conclusion:

Therefore, the number $\sqrt[n]{2}$ is irrational for any integer $n \geq 2$.

B Appendix B. Proof that the limit of the function $f(n) = \sqrt[n]{2 \cdot n}$ is 1

B.1 Problem Statement

Prove that the limit of the function $f(n) = \sqrt[n]{2 \cdot n}$ as n approaches infinity is equal to 1.

B.2 Solution

Consider the function $f(n) = \sqrt[n]{2 \cdot n}$. Our goal is to find the limit of this function as $n \rightarrow \infty$. Let's rewrite the function in the form of a power:

$$f(n) = (2 \cdot n)^{\frac{1}{n}}$$

To find the limit of this function, let's consider the natural logarithm of $f(n)$:

$$\ln(f(n)) = \ln\left((2 \cdot n)^{\frac{1}{n}}\right)$$

Using the properties of logarithms, we get:

$$\ln(f(n)) = \frac{1}{n} \ln(2 \cdot n)$$

Separate the logarithm of the product into the sum of logarithms:

$$\ln(f(n)) = \frac{\ln(2) + \ln(n)}{n}$$

Now, let's find the limit of $\ln(f(n))$ as $n \rightarrow \infty$:

$$\lim_{n \rightarrow \infty} \ln(f(n)) = \lim_{n \rightarrow \infty} \frac{\ln(2) + \ln(n)}{n}$$

This limit has the indeterminate form $\frac{\infty}{\infty}$, so we can apply L'Hôpital's Rule. Take the derivatives of the numerator and denominator with respect to n :

$$\frac{d}{dn}(\ln(2) + \ln(n)) = \frac{1}{n}$$

$$\frac{d}{dn}(n) = 1$$

Thus, the limit becomes:

$$\lim_{n \rightarrow \infty} \frac{\frac{1}{n}}{1} = \lim_{n \rightarrow \infty} \frac{1}{n} = 0$$

So, we have found that the limit of the natural logarithm of the function is 0:

$$\lim_{n \rightarrow \infty} \ln(f(n)) = 0$$

Now, to find the limit of the original function $f(n)$, we use the continuity of the exponential function:

$$\lim_{n \rightarrow \infty} f(n) = \lim_{n \rightarrow \infty} e^{\ln(f(n))} = e^{\lim_{n \rightarrow \infty} \ln(f(n))} = e^0 = 1$$

B.3 Conclusion

Therefore, we have proven that the limit of the function $f(n) = \sqrt[n]{2 \cdot n}$ as n approaches infinity is equal to 1.

C Appendix C. A short proof of monotonicity

It is well known that $\sqrt[n]{2 \cdot n} \rightarrow 1$ as $n \rightarrow \infty$ (see Appendix B). However, in this appendix we focus on showing that $(2 \cdot n)^{1/n}$ is strictly decreasing for $n \geq 3$. Below is one way to prove this, using the logarithmic derivative:

C.1 Define the function.

Let

$$f(n) = (2 \cdot n)^{\frac{1}{n}}.$$

It is convenient to work with its natural logarithm:

$$\ln(f(n)) = \ln((2 \cdot n)^{\frac{1}{n}}) = \frac{1}{n} \ln(2 \cdot n).$$

Denote

$$g(n) = \ln(f(n)) = \frac{\ln(2 \cdot n)}{n} = \frac{\ln(2) + \ln(n)}{n}.$$

C.2 Compute the derivative.

Treating n as a real variable $n > 0$, we have

$$g'(n) = \frac{d}{dn} \left(\frac{\ln(2 \cdot n)}{n} \right).$$

Using the quotient rule,

$$g'(n) = \frac{\frac{d}{dn}[\ln(2 \cdot n)] \cdot n - \ln(2 \cdot n) \cdot 1}{n^2} = \frac{\frac{1}{n} \cdot n - \ln(2 \cdot n)}{n^2} = \frac{1 - \ln(2 \cdot n)}{n^2}.$$

(Note that $\frac{d}{dn}[\ln(2 \cdot n)] = \frac{1}{2 \cdot n} \cdot 2 = \frac{1}{n}$.)

C.3 Sign of the derivative.

We want to see where $g'(n) < 0$:

$$g'(n) < 0 \iff 1 - \ln(2 \cdot n) < 0 \iff \ln(2 \cdot n) > 1 \iff 2 \cdot n > e \iff n > \frac{e}{2}.$$

Since $e \approx 2.718$, the inequality $n > e/2$ is certainly true for all integer $n \geq 2$, and hence strictly for $n \geq 3$. Therefore, for $n \geq 3$, $g(n)$ is strictly decreasing.

C.4 Implication for $f(n)$.

Since $f(n) = \exp(g(n))$, and $\exp(\cdot)$ is a strictly increasing function in its argument, $f(n)$ decreases whenever $g(n)$ decreases. Hence, $f(n)$ is indeed strictly decreasing for $n \geq 3$.

Thus, only for $n = 1$ and $n = 2$ does $f(n)$ attain the maximum value 2, and for all $n \geq 3$, it strictly decreases (which is consistent with the limit $\lim_{n \rightarrow \infty} (2 \cdot n)^{1/n} = 1$).

C.5 Conclusion

Hence, we have demonstrated that $f(n)$ is strictly decreasing for $n \geq 3$, either by analyzing the derivative of $\ln(f(n))$ (as shown above) or by comparing $f(n+1)$ and $f(n)$. This completes the proof of the monotonicity of $f(n)$.

D Appendix D: Analysis of Functions and Heuristic Visualization of Critical Points

D.1 Introduction

The goal of this analysis is to study the behavior of the auxiliary continuous function

$$f(p, q) = q - \sqrt[p]{qp},$$

and to identify its critical points, in particular the points where the second derivative $f''(p)$ equals zero.

This study is not intended as an independent proof of Fermat's Last Theorem. Rather, it provides a geometric and analytic visualization of the special role of the binary/quadratic point

$$p = 2, \quad q = 2$$

in relation to the coefficient-symmetry compatibility scheme discussed in the main article.

In the main text, the decisive algebraic comparison is expressed in terms of two positive real residual scales l and q , arising from two natural normalizations of the same odd binomial core:

$$S_n(m, p) = n l^n, \quad S_n(m, p) = 2^{n-1} q^n.$$

Equivalently, for the full difference of powers,

$$y^n = 2n l^n, \quad y^n = 2^n q^n.$$

Comparing these two normalizations gives the exact compatibility relation

$$l^n = q^n \frac{2^{n-1}}{n}.$$

Therefore, within these two normalizations, the equality of the two residual scales is equivalent to the coefficient-mass equality

$$l^n = q^n \iff n = 2^{n-1}.$$

Since the equation $n = 2^{n-1}$ has only the positive integer solutions

$$n = 1, \quad n = 2,$$

the equality of the two residual scales is compatible with the binomial coefficient structure only in the linear and quadratic cases, and necessarily fails for every exponent $n > 2$.

In this appendix, the symbol q denotes an auxiliary positive real parameter in the function $f(p, q)$. It should not be confused with a modulus q used in modular arithmetic or with the finite-field notation $\mathbb{F}_q$. It should also be distinguished from the residual scale q in the main algebraic text, although the same letter is used because both quantities are intended to visualize a positive scaling value. The auxiliary value $q = 2$ is interpreted here only as a geometric marker of the binary/quadratic case, not as an additional axiom and not as a replacement for the algebraic compatibility relation.

Thus, the role of this appendix is heuristic and illustrative. It visualizes why the point $p = 2$, $q = 2$ is geometrically distinguished, while the rigorous algebraic obstruction in the article remains the compatibility chain

$$l^n = q^n \iff n = 2^{n-1} \implies n \in \{1, 2\}.$$

D.2 Results of the Analysis

D.2.1 Combined Graph of Functions

The following graph (Fig. 1) illustrates three functions, each with its corresponding discrete values:

- $1.5 - \sqrt[3]{1.5p}$ (blue line) and $1.5 - \sqrt[3]{1.5n}$ (blue squares),
- $2 - \sqrt[3]{2p}$ (green line) and $2 - \sqrt[3]{2n}$ (green squares),
- $3 - \sqrt[3]{3p}$ (orange line) and $3 - \sqrt[3]{3n}$ (orange squares).

These curves should be read as a visual comparison between continuous values of p and the discrete exponent values n . The case $q = 2$ is singled out because it corresponds to the binary value appearing in the equivalent exponential-linear relation

$$n = 2^{n-1} \quad \Longleftrightarrow \quad 2^n = 2n.$$

In the main text, this relation is not introduced by the graph itself. It follows from the comparison of the two binomial normalizations:

$$S_n(m, p) = n l^n, \quad S_n(m, p) = 2^{n-1} q^n,$$

together with the residual-scale equality $l^n = q^n$. In Version 15 this is expressed by the sharper compatibility statement

$$l^n = q^n \quad \Longleftrightarrow \quad n = 2^{n-1}.$$

The graph does not prove Fermat's Last Theorem and does not prove this algebraic equivalence; it only visualizes the same quadratic compatibility point.

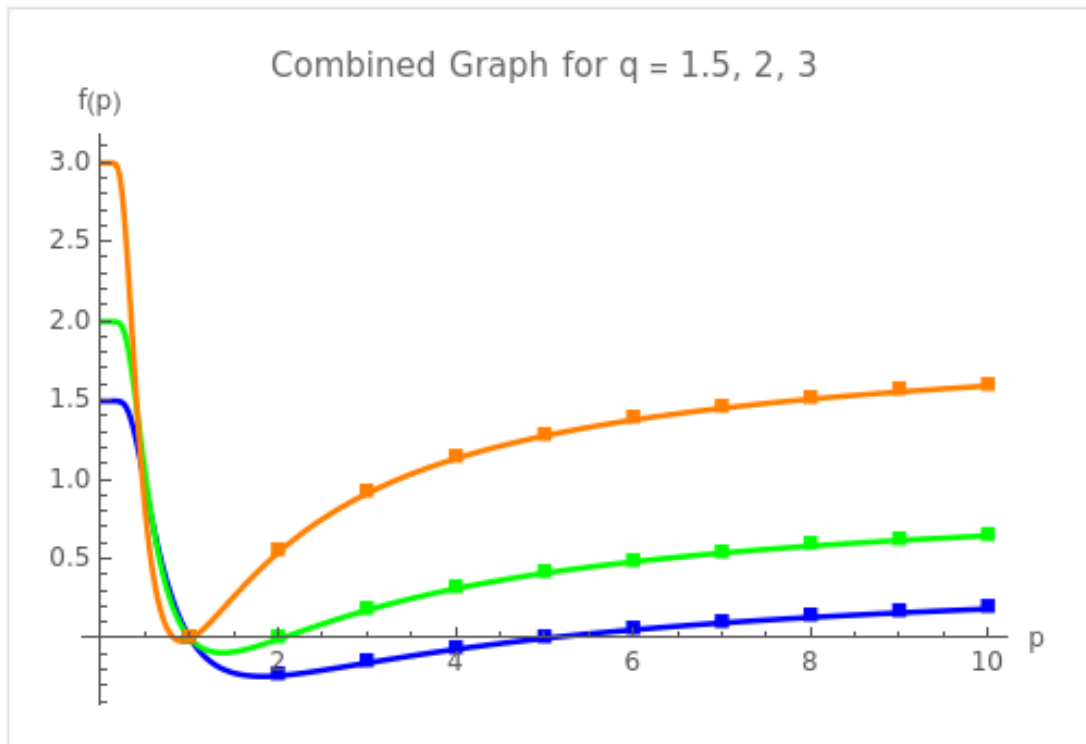


Figure 1: Combined graph of $q - \sqrt[3]{qp}$ for $q = 1.5, 2, 3$. Lines represent continuous p , and squares represent discrete n . The value $q = 2$ marks the binary/quadratic alignment.

D.2.2 Surface Plot of $f(p, q)$

The three-dimensional surface plot below (Fig. 2) illustrates the behavior of

$$f(p, q) = q - \sqrt[p]{qp}$$

for continuous values of $p > 0$ and $q > 0$. The perspective highlights the change in curvature as both parameters vary.

This visualization is meant to support the geometric intuition behind the binary value $q = 2$. The algebraic argument itself rests on the comparison of the two residual scales l and q :

$$y^n = 2n l^n, \quad y^n = 2^n q^n.$$

This comparison gives

$$l^n = q^n \frac{2^{n-1}}{n},$$

and hence the residual-scale equality is compatible with the two binomial normalizations if and only if

$$l^n = q^n \iff n = 2^{n-1}.$$

Therefore, the surface plot is only a heuristic visual aid. It does not prove Fermat's Last Theorem and does not replace the algebraic coefficient-symmetry compatibility obstruction.

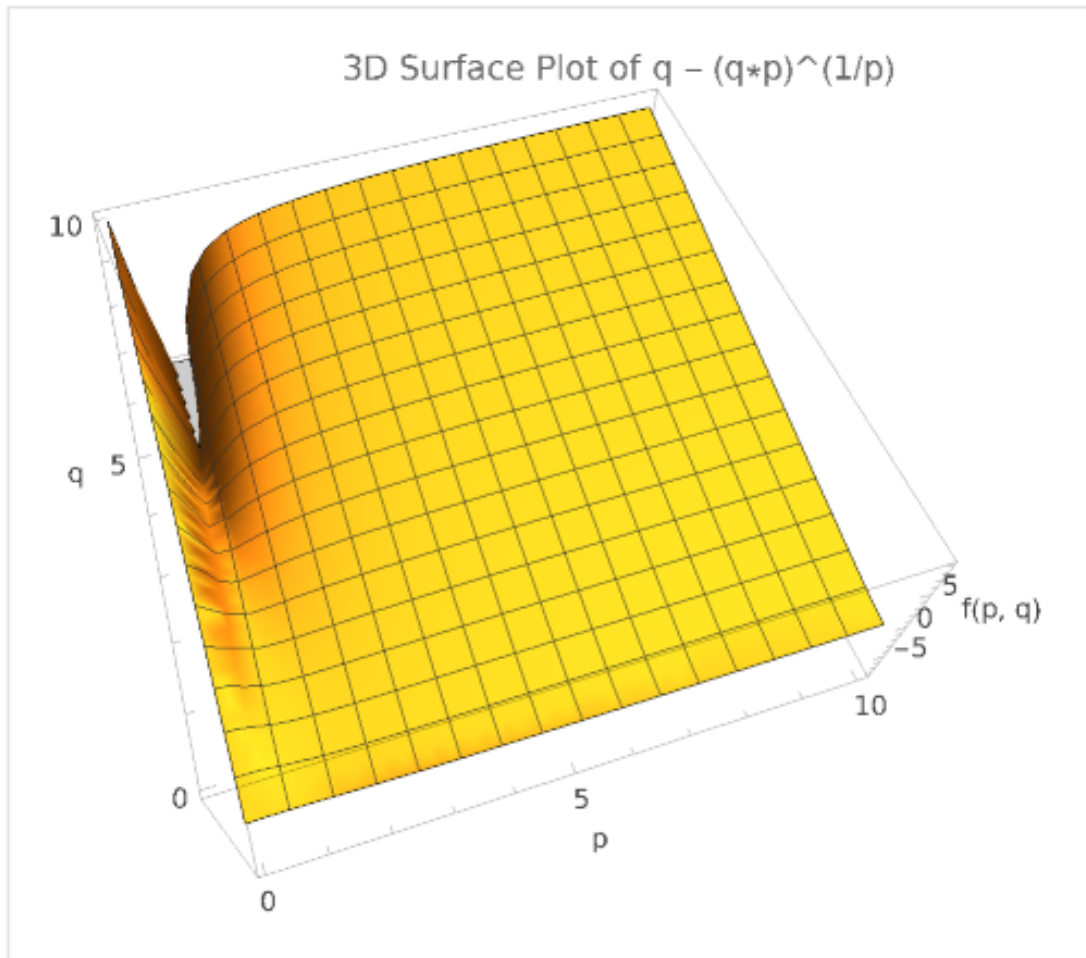


Figure 2: 3D surface plot of $f(p, q) = q - \sqrt[p]{qp}$ for $p > 0$ and $q > 0$. The point $p = 2$, $q = 2$ is interpreted as a geometric marker of the quadratic alignment.

D.3 Analysis and Conclusions

The numerical and graphical analysis suggests the following interpretation:

- The critical point at $p = 2$ appears naturally for $q = 2$, where the second derivative $f''(p)$ equals zero. This highlights the special geometric role of $q = 2$, corresponding to the binary/quadratic alignment.
- For other values of q , critical points may exist, but they occur at values $p \neq 2$. This illustrates that $q = 2$ is distinguished by its alignment with the quadratic case $n = 2$, which is precisely the case of Pythagorean triples.
- In the terminology of the main text, this does not mean that a scaling value is imposed by the graph. Rather, it gives a geometric picture compatible with the algebraic fact that

$$l^n = q^n \iff n = 2^{n-1},$$

and that this compatibility occurs only for $n = 1$ and $n = 2$.

These results should be understood as a geometric consistency check for the algebraic argument of the article. They illustrate the same special role of the binary value

$$2,$$

but they do not constitute an independent proof of Fermat's Last Theorem. The rigorous algebraic part of the argument remains the residual-scale compatibility obstruction

$$l^n = q^n \iff n = 2^{n-1},$$

together with the elementary, and machine-checkable, fact that

$$n = 2^{n-1} \implies n \in \{1, 2\}.$$

E Appendix E: Geometric Motivation for the Coefficient-Symmetry Compatibility Obstruction through the Function $f(p, q)$

In this appendix, we further investigate the connection between the auxiliary function

$$f(p, q) = q - \sqrt[p]{qp}$$

and the scale comparison associated with Fermat's equation

$$x^n + y^n = z^n.$$

The purpose of this appendix is not to provide a standalone geometric proof of Fermat's Last Theorem, but to visualize why the binary/quadratic point

$$p = 2, \quad q = 2$$

is geometrically distinguished.

In the coefficient-symmetry compatibility scheme of the main article, the decisive algebraic step is not the graph itself. The decisive step is the comparison of two real normalizations of the same odd binomial core:

$$S_n(m, p) = n l^n, \quad S_n(m, p) = 2^{n-1} q^n.$$

Equivalently, for the full difference of powers,

$$y^n = 2n l^n, \quad y^n = 2^n q^n.$$

Their comparison gives

$$l^n = q^n \frac{2^{n-1}}{n}.$$

Under these two normalizations of the same odd binomial core, the equality of the two residual scales is therefore equivalent to

$$n = 2^{n-1}.$$

More explicitly,

$$l^n = q^n \iff n = 2^{n-1}.$$

Thus, the final obstruction is the elementary exponential-linear equation

$$n = 2^{n-1},$$

not a graphical property of $f(p, q)$.

The function $f(p, q)$ gives a visual way to see why the point $p = 2, q = 2$ plays a special role: it marks the place where the binary value aligns with the quadratic case, that is, with the existence of Pythagorean triples. In this sense the function supports the geometric intuition behind the coefficient-symmetry compatibility obstruction, but it does not prove Fermat's Last Theorem and does not replace the algebraic comparison.

E.1 Analysis of the Second Derivative

The function

$$f(p, q) = q - \sqrt[p]{qp}$$

allows us to study the behavior of inflection points, which are determined by the condition

$$f_p''(p, q) = \frac{\partial^2}{\partial p^2} \left(q - \sqrt[n]{qp} \right) = 0.$$

Inflection points are useful here because they reveal changes in the shape of the auxiliary function. In the present context, these changes are interpreted as a geometric reflection of the transition from the quadratic case $n = 2$ to higher exponents $n > 2$.

The notation p in this appendix is used as a continuous variable, comparable with the discrete exponent n . It should not be confused with the parameter p in the main parametrization

$$z = m^n + p^n, \quad x = m^n - p^n.$$

Likewise, the notation q in this appendix is a positive real visualization parameter; it is not a modulus and is not a finite-field parameter.

E.2 Additional Analysis for the Variable q

Investigations were also conducted on the variable q . At the point $q = 2$, with $p = 2$, both the first and second partial derivatives with respect to q are equal to 0.5. This result indicates a special local behavior of the auxiliary function at the binary/quadratic point.

However, in the main argument of the article, the value $q = 2$ is not introduced as a consequence of this numerical observation. Rather, the graph should be read only as an additional geometric illustration of why the quadratic alignment

$$p = 2, \quad q = 2$$

is visually distinguished.

Thus, the differential analysis in this appendix should be read as an auxiliary geometric commentary, not as a replacement for the algebraic scale comparison

$$y^n = 2n l^n, \quad y^n = 2^n q^n,$$

nor as a proof of Fermat's Last Theorem. Its role is only to visualize the same compatibility obstruction:

$$l^n = q^n \iff n = 2^{n-1}.$$

E.3 Numerical Results

Below are the numerical values of the second derivative $f_p''(p, q)$ for various values of p and q :

p	$q = 1.5$	$q = 2.0$	$q = 2.5$	$q = 3.0$	$q = 4.0$
1	2.75	0.77	2.90	0.96	1.10
2	0.17	0.00	-0.11	-0.34	-0.55
3	-0.0056	-0.08	-0.099	-0.21	-0.31
4	-0.018	-0.11	-0.056	-0.12	-0.17
5	-0.014	-0.10	-0.033	-0.067	-0.093

Table 2: Numerical values of the second derivative $f_p''(p, q)$ for various p and q .

The table shows that the pair

$$p = 2, \quad q = 2$$

is singled out by the vanishing of the second derivative in the numerical data. This is consistent with the special role of the quadratic case, but it should not be interpreted as a direct proof that higher-degree integer solutions are impossible.

It is important to emphasize that this table is not used to derive the equivalence

$$l^n = q^n \iff n = 2^{n-1}.$$

This equivalence comes from the algebraic comparison of the two normalizations of the same odd binomial core. The numerical table merely visualizes why the point associated with the quadratic case is geometrically distinguished.

E.4 Conclusions from the Data

- For $p = 2, q = 2$ we have $f_p''(p, q) = 0$, which geometrically corresponds to the quadratic case and is consistent with the existence of Pythagorean triples.
- For $q > 2$, the inflection point shifts toward lower values of p . This suggests a change in the shape of the auxiliary function and is consistent with the loss of the special quadratic alignment.
- Figure 3 illustrates the surface of $f_p''(p, q)$ with the marked inflection points.
- This geometric observation supports the interpretation that the binary/quadratic point $p = 2, q = 2$ is special, but it does not by itself prove Fermat's Last Theorem or the algebraic compatibility equivalence.

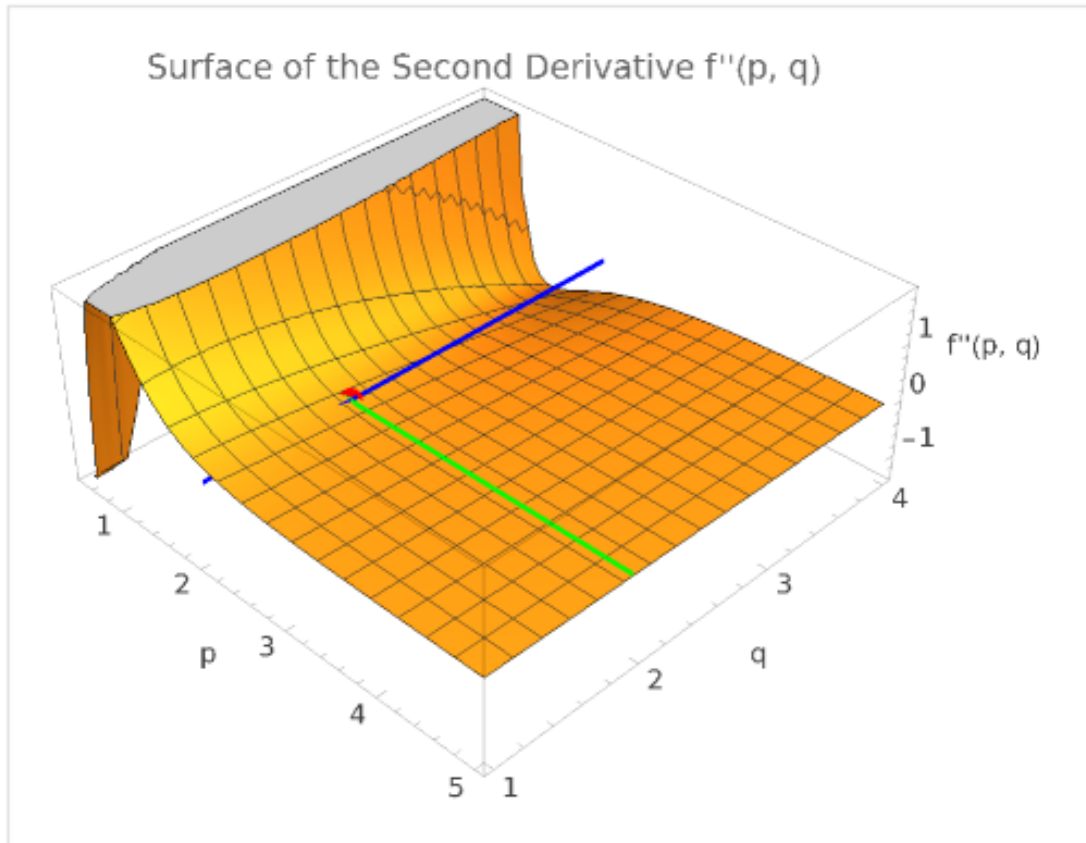


Figure 3: Surface of the second derivative $f_p''(p, q)$, demonstrating the shift of the inflection point and the special role of the point $p = 2, q = 2$.

E.5 Geometric Interpretation in Relation to Fermat's Equation

E.5.1 Relation to the Inflection Points

From the data, one can draw the following heuristic interpretation:

1. If the inflection point is aligned with the discrete exponent value $p = n$, then the auxiliary geometry is compatible with the corresponding exponent.
2. The alignment $p = 2$, $q = 2$ is consistent with the quadratic case $n = 2$, where integer solutions exist in the form of Pythagorean triples.
3. For $q > 2$, the shift of the inflection point away from $p = 2$ illustrates the loss of this quadratic alignment.
4. This loss of geometric alignment is compatible with, but does not prove, the algebraic obstruction obtained from the coefficient-symmetry compatibility relation.

Thus, the geometric picture distinguishes the quadratic case

$$n = 2,$$

while the exclusion of exponents

$$n > 2$$

comes from the algebraic compatibility relation

$$l^n = q^n \iff n = 2^{n-1} \implies n \in \{1, 2\}.$$

E.5.2 Final Conclusion

Based on the numerical and graphical calculations:

- **For $q = 2$, the inflection point corresponds to $p = 2 \Rightarrow$** the auxiliary geometry is aligned with the quadratic case, where Fermat's equation has integer solutions through Pythagorean triples.
- **For $q > 2$, the inflection point shifts to lower values of $p \Rightarrow$** the auxiliary geometry no longer has the same alignment with the quadratic case.
- This supports the interpretation that the binary/quadratic point

$$p = 2, \quad q = 2$$

is geometrically distinguished in the coefficient-symmetry compatibility framework of the article.

Thus, the function $f(p, q)$ provides a useful geometric visualization of the special role of the binary value $q = 2$ and of the quadratic exponent $p = 2$. It supports and illustrates the main algebraic argument, but the strict exclusion of positive integer solutions for

$$x^n + y^n = z^n, \quad n > 2,$$

does not rest on a graphical property. It rests on the algebraic comparison

$$y^n = 2n l^n, \quad y^n = 2^n q^n,$$

which gives the residual-scale compatibility relation

$$l^n = q^n \iff n = 2^{n-1}.$$

Since this equation has only the positive integer solutions

$$n \in \{1, 2\},$$

the residual-scale equality is aligned with the quadratic case and is incompatible with all exponents higher than the square.